1. Introduction, Examples and Applications

Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.3 Forecasting competitions and prediction markets

Forecasting

"It's difficult to make predictions, especially about the future."

Danish proverb, frequently attributed to Niels Bohr.

A major human desire is to make forecasts for an uncertain future.

Consequently, forecasts ought to be probabilistic in nature, taking the form of probability distributions over future quantities or events.

Over the past decade(s), a transition from point prediction to distributional prediction, or probabilistic forecasting, can be observed across applications.

Forecasting

- Suppose we are trying to predict $Y \in \mathcal{Y}$
  - Binary outcomes: $Y \in \{0, 1\}$
  - Categorical outcomes: $Y \in \{1, 2, \ldots, m\}$
  - Univariate outcomes: $Y \in \mathbb{R}$
  - Multivariate outcomes: $Y \in \mathbb{R}^d$
- Predictions come in many forms
  - A point prediction is a single value $x \in \mathcal{Y}$
  - A probabilistic prediction is a probability distribution F over $\mathcal{Y}$
  - A prediction interval (or prediction set) is a set $I \subset \mathcal{Y}$
- We will focus mainly on probabilistic predictions
  - Point predictions can be derived from probabilistic predictions
  - e.g., by taking the mean, median, or a quantile
  - Prediction intervals can also be derived from probabilistic predictions

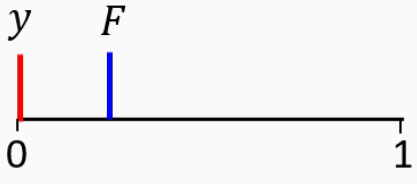
Forecasting

- Example: Binary outcomes



(a) Point forecast

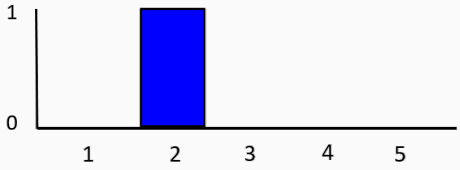
$x \in \{0, 1\}$



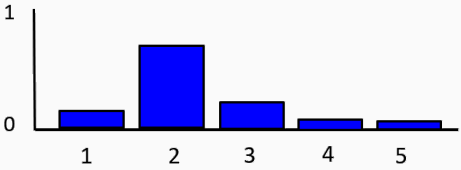
(b) Probabilistic forecast

$F \in [0, 1]$

- Example: Categorical outcomes



(a) Point forecast

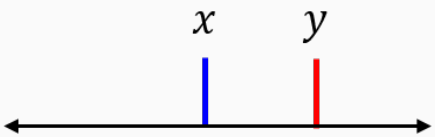


(b) Probabilistic forecast

$x \in \{1, 2, \ldots, m\}$

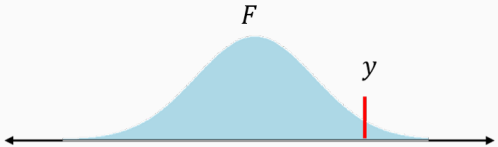
$F \in [0, 1]^m \quad \text{with} \quad \sum_{j=1}^{m} F_j = 1$

- Example: Univariate outcomes



(a) Point forecast

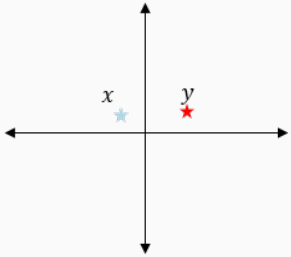
$x \in \mathbb{R}$



(b) Probabilistic forecast

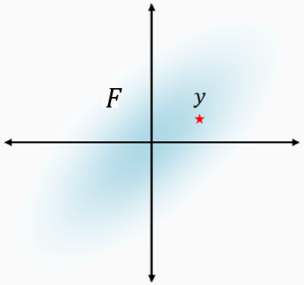
F is a probability distribution on $\mathbb{R}$

- Example: Multivariate outcomes



(a) Point forecast

$x \in \mathbb{R}^d$



(b) Probabilistic forecast

F is a probability distribution on $\mathbb{R}^d$

Probabilistic vs. point predictions

We make predictions to obtain useful information about the future that we can use to aid decision making and risk assessment.

Probabilistic predictions quantify our uncertainty and are more useful for decision making.

Point forecasts are typically easier to communicate, and most applications require "deterministic" decisions:

- cancel an event or not
- build a dam with a specific height
- sell or hold a position in an investment portfolio
- operate a water reservoir
- ...

Learning goals

The overarching aim of the lecture is to learn how to make and evaluate probabilistic forecasts.

We will cover

- how probabilistic forecasts are used across applications, from economics to weather and climate
- mathematical foundations of how probabilistic (and point) forecasts can be evaluated
- how we can design, implement and estimate statistical and machine learning methods that yield probabilistic forecasts

Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.3 Forecasting competitions and prediction markets

Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.2.1 Economic forecasting

1.2.2 Political and social science

1.2.3 Sports analytics

1.2.4 Health and epidemiology

1.2.5 Energy forecasting

1.2.6 Weather and climate

1.2.7 Other environmental sciences applications

1.3 Forecasting competitions and prediction markets

Macroeconomic forecasting

Process of predicting economy-wide performance and major indicators like gross domestic product (GDP), inflation, and unemployment.

Forecasts are often made in quarter-yearly resolution, with time horizons short-term updates to multi-year projections.

Common methods for producing such forecasts (for an overview, see Stock and Watson, 2006) include

- theory-driven structural models
- data-driven statistical models, e.g., time series models
- more recently also machine learning models

We will consider two example, the inflation forecasts by the Bank of England, and the Survey of Professional Forecasters by the European Central Bank.

Bank of England inflation forecasts

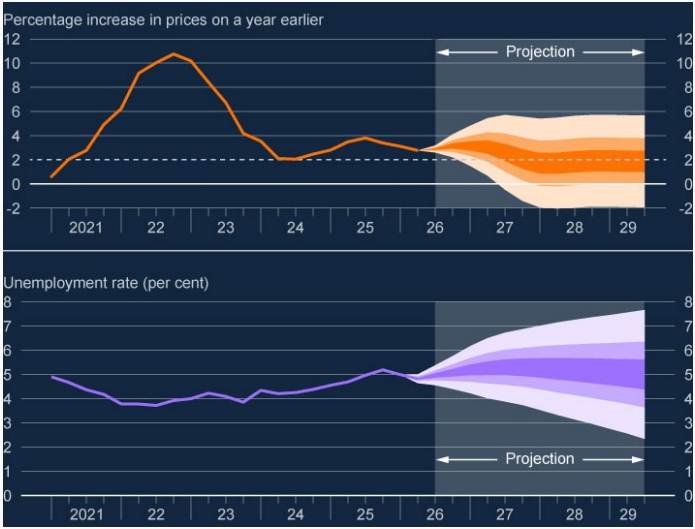
The Bank of England's Monetary Policy Committee (MPC) has issued probabilistic forecasts of inflation rates every quarter since February 1996.

Fan charts are used to visualize selected deciles of the predictive distributions.

The fan charts are constructed using historical forecast errors as the empirical basis, with the MPC judgementally adjusting the width and skew of the distribution.

Current forecasts are available at
`https://www.bankofengland.co.uk/monetary-policy-report/2026/july-2026`.

Bank of England inflation forecasts



https://www.bankofengland.co.uk/monetary-policy-report/2026/july-2026, Section 3.2

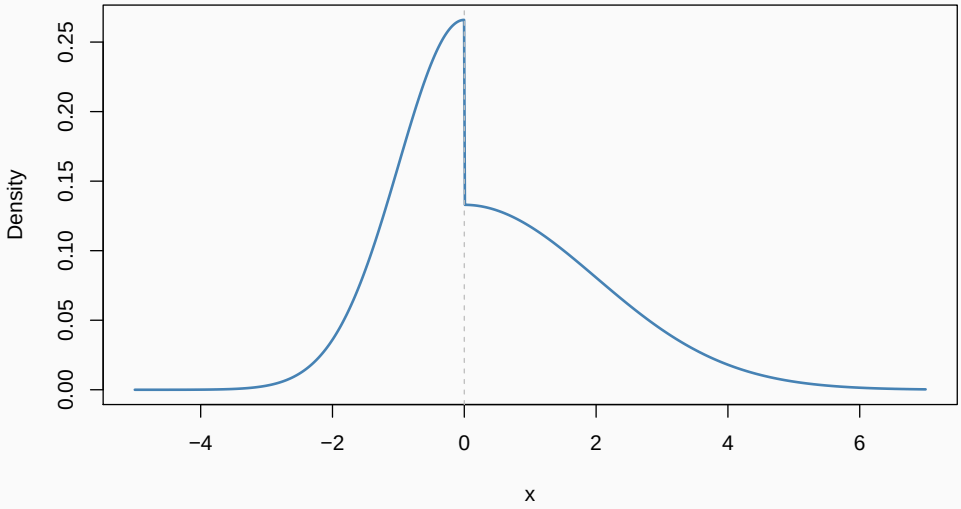
Bank of England inflation forecasts

The MPC employs potentially asymmetric two-piece normal distributions with parameters $\mu \in \mathbb{R}$ and $\sigma_1, \sigma_2 > 0$ and forecast density

$$f(y) = \begin{cases} \left(\frac{\pi}{2}\right)^{-1/2} (\sigma_1 + \sigma_2)^{-1} \exp\left(-\frac{(y-\mu)^2}{2\sigma_1^2}\right) & \text{if} \quad y \leq \mu, \\ \left(\frac{\pi}{2}\right)^{-1/2} (\sigma_1 + \sigma_2)^{-1} \exp\left(-\frac{(y-\mu)^2}{2\sigma_2^2}\right) & \text{if} \quad y \geq \mu. \end{cases} \tag{1.1}$$

In the exercises, you will find the expectation and variance of this density and prove that it is right-skewed when $\sigma_2 > \sigma_1$.

Bank of England inflation forecasts



European Central Bank Survey of Professional Forecasters

There is a long history of systematic surveys of professional economists' macroeconomic forecasts, in particular in the US.

The Survey of Professional Forecasters (SPF) is a quarterly survey conducted by the European Central Bank among professional forecasters, which has been running since 1999.

- conducted in January, April, July and October
- respondents are economists/forecasters working at financial institutions, research institutions and other organizations
- main variables (probabilistic forecasts): inflation, GDP growth, unemployment rate
- secondary variables (point forecasts): wages, USD/EUR exchange rate, oil prices, and others

More details: `https://www.ecb.europa.eu/press/economic-bulletin/html/eb201901~a3afea780e.en.html#IDofArticle1`

European Central Bank Survey of Professional Forecasters

For the main variables, respondents provide a probability distribution over possible outcomes.

An individual response is a point forecast plus a histogram-like subjective predictive distribution, where the bins have a width of 0.5 percentage points, e.g.,

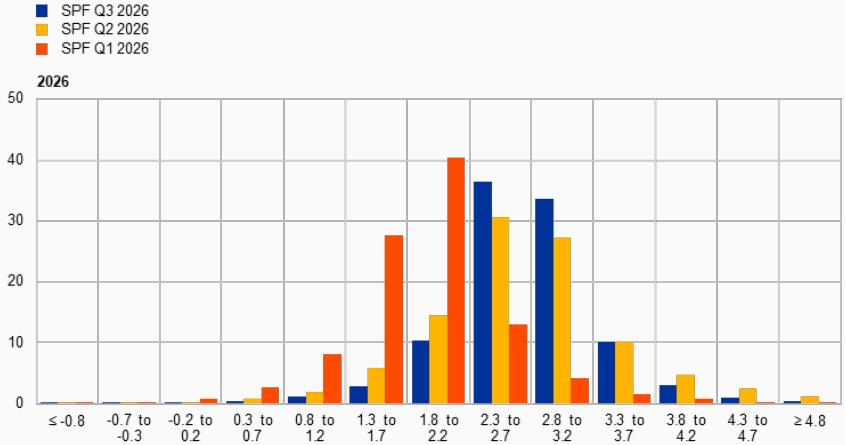
Inflation rate	< 0	$0-0.5$	$0.5-1.0$	$1.0-1.5$	$1.5-2.0$	$2.0-2.5$	$2.5-3.0$	$3.0-3.5$	≥ 3.5
$\mathbb{P}_i(Y \in B_k)$	0.01	0.02	0.04	0.10	0.25	0.32	0.16	0.07	0.03

The ECB aggregates these individual probability assessments into aggregate probability distributions.

Current forecasts are available at `https://www.ecb.europa.eu/stats/ecb_surveys/survey_of_professional_forecasters/html/ecb.spf2026q3.en.html`.

European Central Bank Survey of Professional Forecasters

Aggregate probability distributions for expected inflation in 2026:



https://www.ecb.europa.eu/stats/ecb_surveys/survey_of_professional_forecasters/html/ecb.spf2026q3.en.html

Other economic forecasting applications

Point and probabilistic forecasts are highly relevant in many areas of economics and business, for example

- exchange rate forecasting
- financial forecasting (e.g., stock prices, portfolio management)
- risk management
- supply chain management (demand and production forecasting)
- forecasting new product success
- predictive maintenance

Overview and more applications: Petropoulos et al. (2022).

Example: Risk management

The expected shortfall or α-conditional value-at-risk functional (CVaR_α, $0 < \alpha < 1$) equals the expectation of a random variable X conditional on it taking values in its upper $(1-\alpha)$-tail, i.e.,

$$\mathrm{CVaR}_\alpha(X) = \frac{1}{1-\alpha}\mathbb{E}\left[X \cdot \mathbf{1}_{X \geq q_\alpha(X)}\right],$$

where q_α denotes the α-quantile.

The CVaR a popular risk measure in quantitative finance and plays a key role in the Basel III - regulations of the Basel Committee on Banking Supervision of the Bank for International Settlements (BIS) governing the regulation of banks.

Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.2.1 Economic forecasting

1.2.2 Political and social science

1.2.3 Sports analytics

1.2.4 Health and epidemiology

1.2.5 Energy forecasting

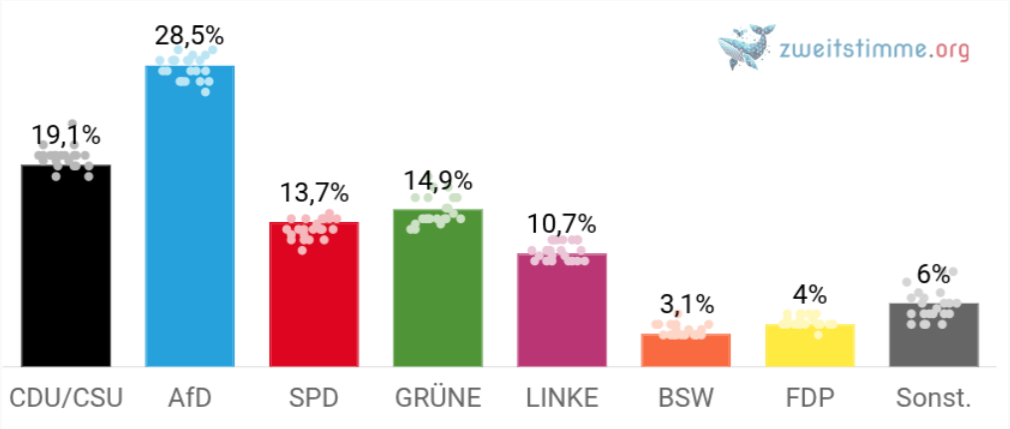
1.2.6 Weather and climate

1.2.7 Other environmental sciences applications

1.3 Forecasting competitions and prediction markets

Election forecasting

Opinion polls are widespread, but can/should not be interpreted as forecasts (and usually offer no probabilistic information).



https://zweitstimme.org/, accessed October 5, 2026

Election forecasting

Some platforms (e.g., FiveThirtyEight (now discontinued), Zweitstimme.org) produce probablistic election forecasts:



https://zweitstimme.org/blog/posts/be-mv-2026-evaluation/,

Probabilistic forecasts for the state parliament election in Mecklenburg-Vorpommern in September 2026

Challenges in political forecasting

- small sample sizes, with, e.g., national elections occurring every 4-5 years only
- data scarcity
- complex, non-linear reactions to events (e.g., economic shocks)
- feedback loops: forecasts may influence the outcomes, e.g., due to
  - voter behavior (e.g., strategic voting)
  - turn-out dynamics
  - donor behavior

Conflict early warning and violent unrest

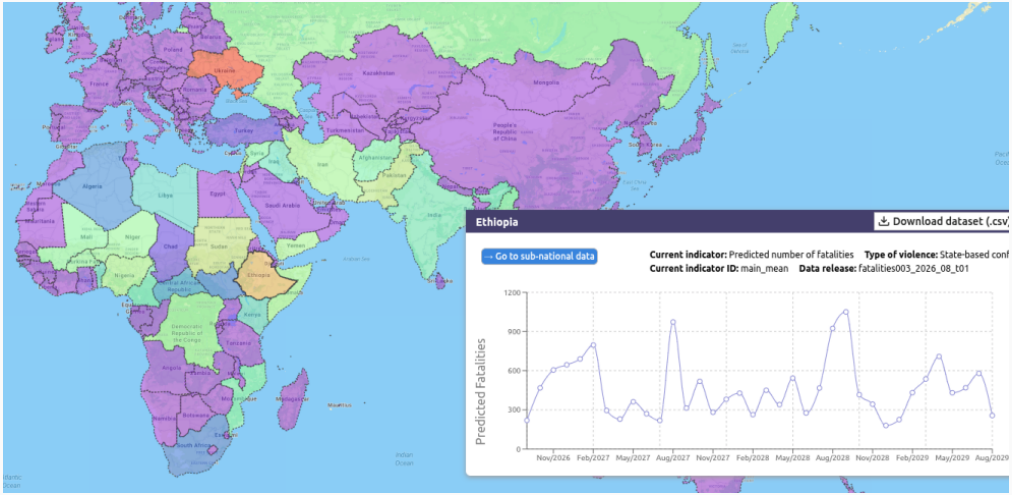
Governments, international agencies, and NGOs utilize (probabilistic) forecasts to allocate humanitarian aid and diplomatic resources.

Often event-based forecasting, i.e., predicting the probability of conflict onset, escalation, or termination in specific grid cells or countries over a 1–12 month horizon.

ViEWS (Violence & Impacts Early-Warning System) Consortium:

- runs "prediction challenges" and produces deterministic and probabilistic forecasts, e.g., of the number of fatilities in armed conflict
- platform collecting forecast and observation data
- details: Hegre et al. (2019)

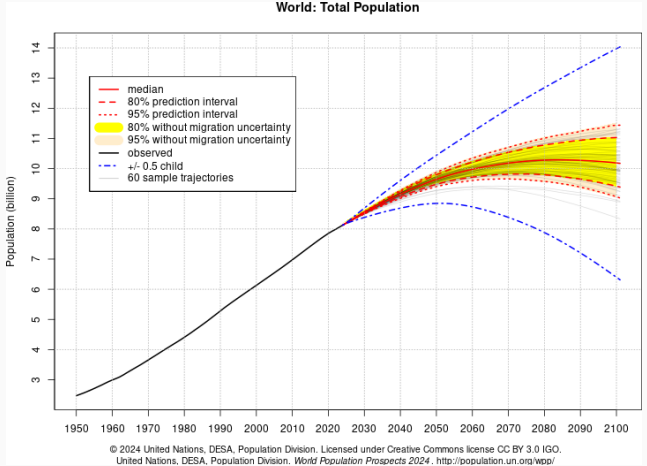
Conflict early warning and violent unrest



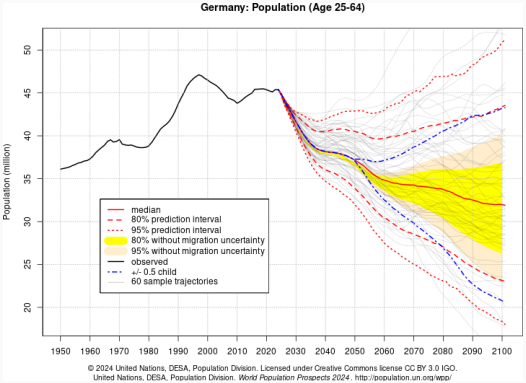
https://data.viewsforecasting.org/

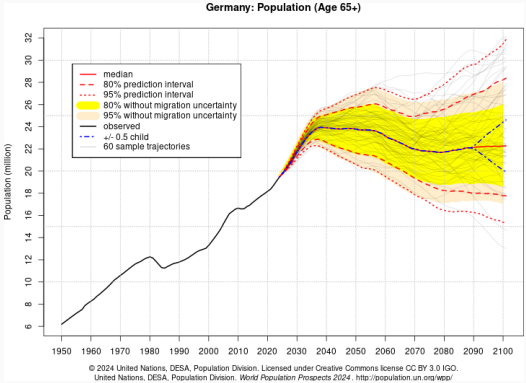
Demography

Probabilistic population predictions are available from the United Nations. Details on methodology: Raftery et al. (2014)



UN population predictions for Germany





Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.2.1 Economic forecasting

1.2.2 Political and social science

1.2.3 Sports analytics

1.2.4 Health and epidemiology

1.2.5 Energy forecasting

1.2.6 Weather and climate

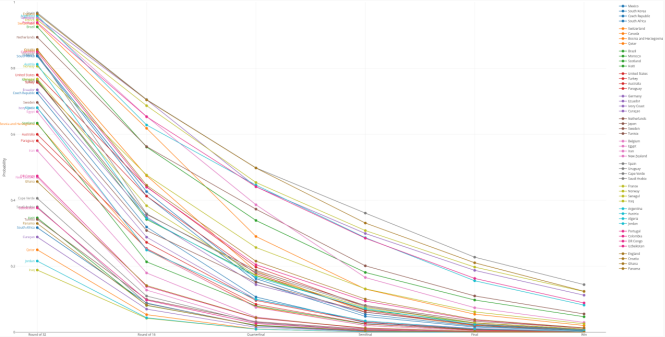
1.2.7 Other environmental sciences applications

1.3 Forecasting competitions and prediction markets

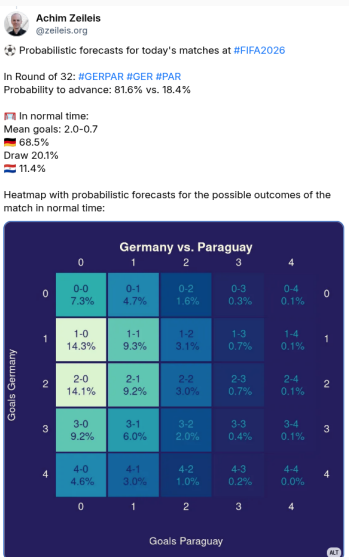
Match- and tournament-level prediction

Probabilistic modeling of match-level outcomes is of interest to betting markets, team management, and fan media.

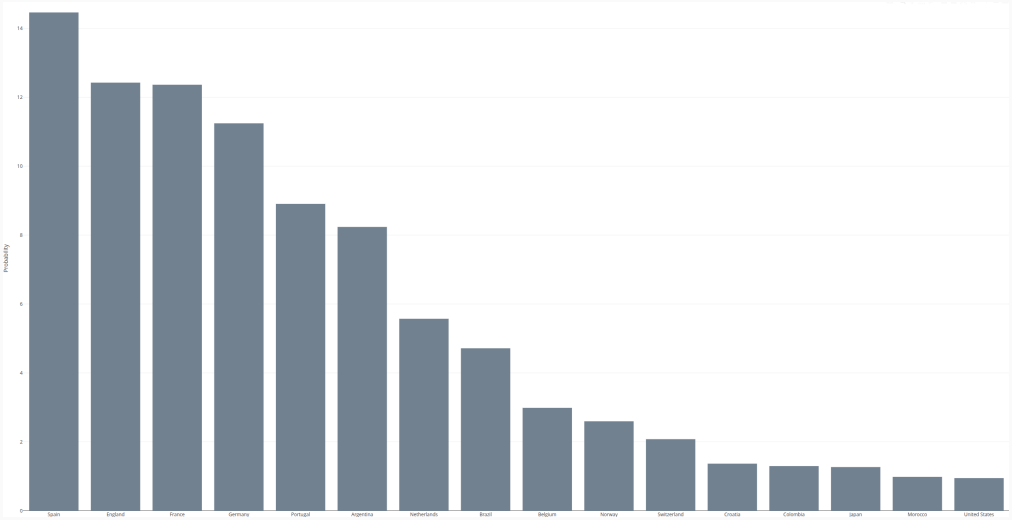
Simulated games based on estimated winning probabilities enable tournament-level predictions.



https://www.zeileis.org/news/fifa2026/#winning-probabilities



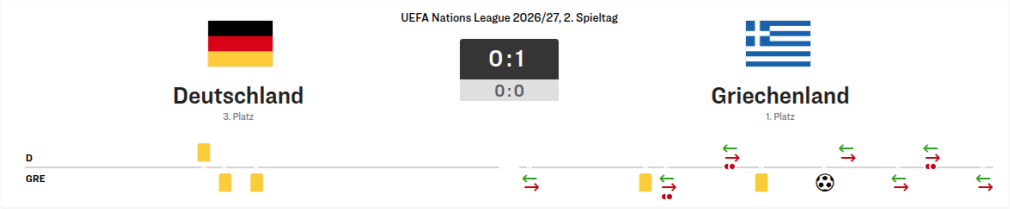
Match- and tournament-level prediction



https://www.zeileis.org/news/fifa2026/#winning-probabilities

Event-level probabilities

Tracking data allows to evaluate individual situations, e.g., estimates of the probability of a goal given the shot location, defender positions, etc. to compute a team's expected goals



https://www.kicker.de/griechenland-gegen-deutschland-2026-uefa-nations-league-5193276/spieldaten

Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.2.1 Economic forecasting

1.2.2 Political and social science

1.2.3 Sports analytics

1.2.4 Health and epidemiology

1.2.5 Energy forecasting

1.2.6 Weather and climate

1.2.7 Other environmental sciences applications

1.3 Forecasting competitions and prediction markets

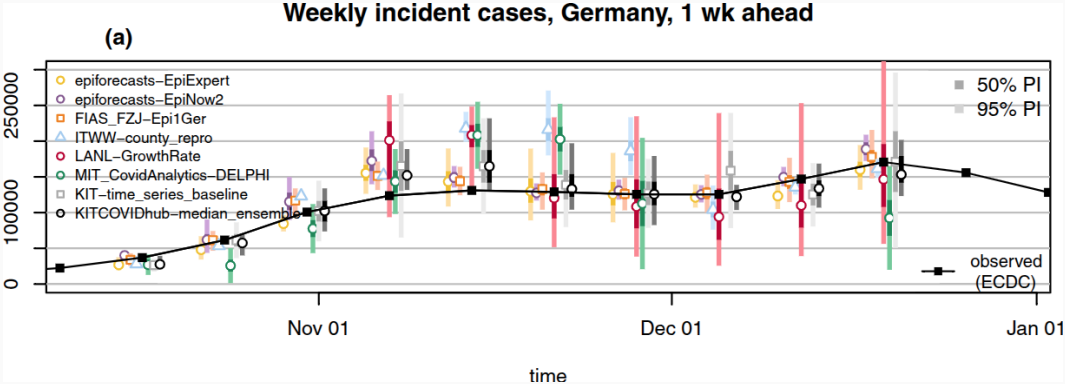
Health forecasting

Important health domain applications include

- epidemiology:
  - targets: incident cases, reproduction number (R_t), deaths
  - usually on the level of regions or entire countries
  - highly relevant during the COVID-19 pandemic for political response and management
- operational healthcare forecasting at hospitals, intensive care units (ICUs), etc.
  - targets: occupancy, length of stay (LoS), arrivals
  - patient-level risks and outcomes
  - increasingly informed by epidemiological modeling (e.g. during the COVID-19 pandemic)

Example: COVID-19 forecasting hubs

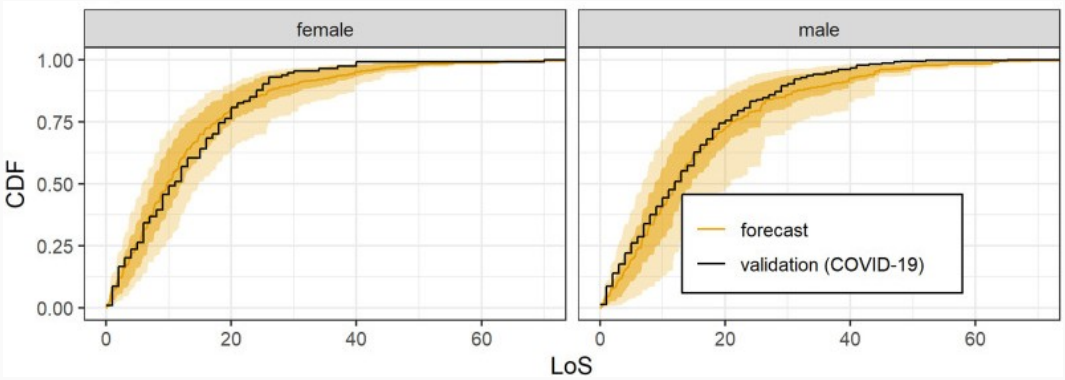
Starting with the US, forecasting hubs became available as open comparison platform for teams to submit probabilistic, real-time model predictions.



Details: Bracher et al. (2021)

Example: Intensive care unit (ICU) length of stay (LoS)

Probabilistic LoS prediction of patients in Swiss ICUs with the aim of improving resource allocation and patient care during the COVID-19 pandemic:



Details: Henzi et al. (2021)

Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.2.1 Economic forecasting

1.2.2 Political and social science

1.2.3 Sports analytics

1.2.4 Health and epidemiology

1.2.5 Energy forecasting

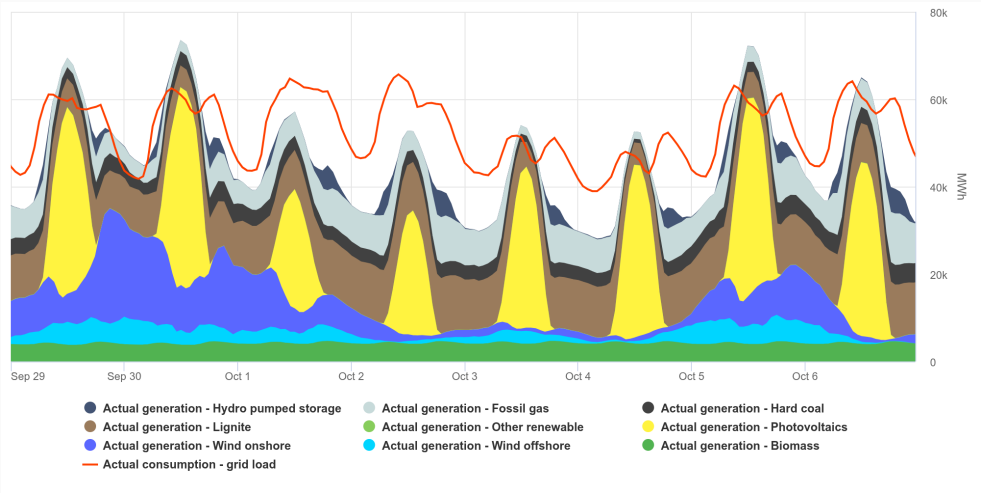
1.2.6 Weather and climate

1.2.7 Other environmental sciences applications

1.3 Forecasting competitions and prediction markets

Energy generation in Germany

Energy generation in Germany is highly weather-dependent, with high variability



https://www.smard.de/en/marktdaten/

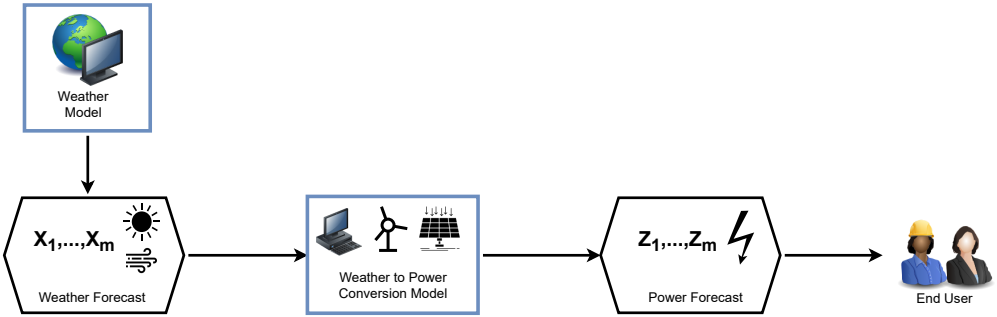
Energy forecasting

Energy forecasting is relevant for several applications, including

- renewable energy generation (in particular wind and solar energy)
- electric load (i.e., consumption)
- electricity prices
- power system planning (e.g., battery storage) and operation (e.g., maintenance)

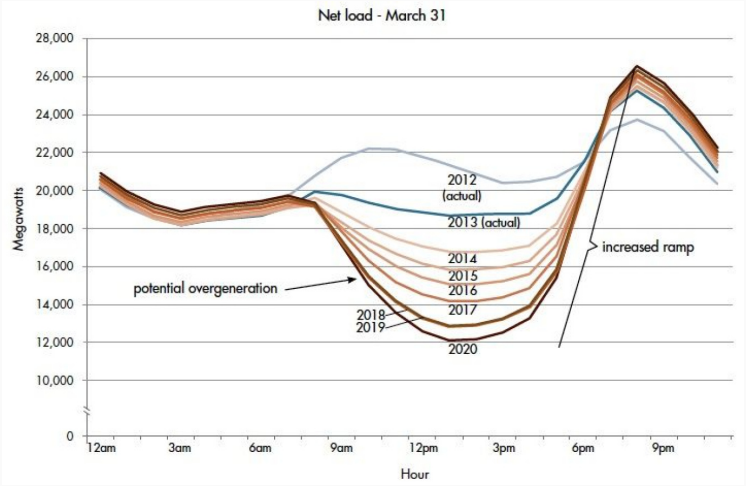
Overview: Hong et al. (2020)

Renewable energy generation forecasting



Electric load forecasting

"Duck curve effect": Net load (i.e., difference of actual load and solar energy generation) in California:



https://www.energy.gov/cmei/solar/

Electricity price forecasting

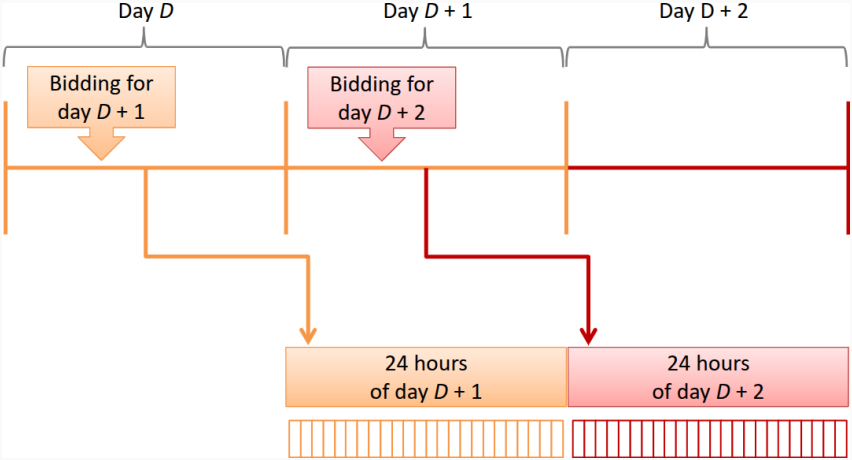
Electricity can be traded on various wholesale spot markets in Europe



https://www.epexspot.com/en/basicspowermarket

Day-ahead electricity price forecasting

The day-ahead market is of particular importance



Electricity price forecasting (EPF)

EPF is important because it cannot be stored at scale easily or cheaply, and supply and demand must match continuously in real time across an interconnected physical grid.

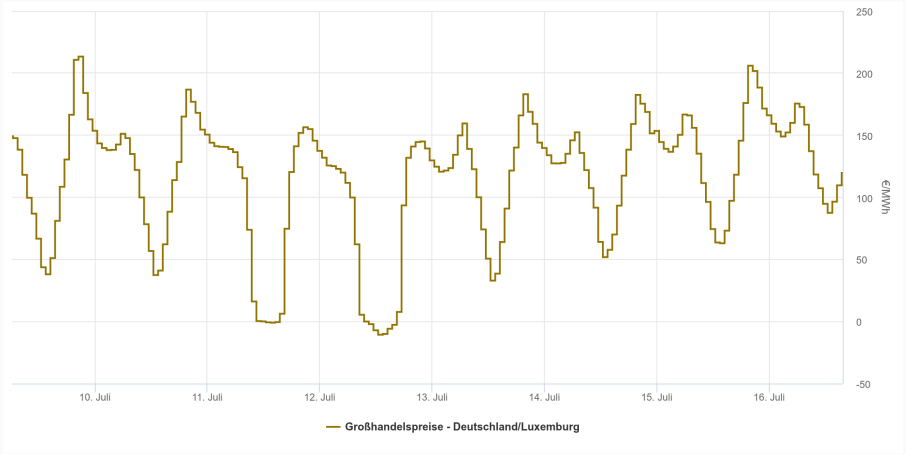
Probabilistic forecasts are particularly important for risk management.

Market participants include

- Generators of electricity, who need to decide whether to clear their marginal generation costs or withhold capacity, e.g. for scheduling maintenance or avoiding negative prices
- Retailers & utilities, who need to build competitive fixed- or dynamic-rate retail tariffs for consumers while covering market risks
- Proprietary traders

Electricity price forecasting (EPF)

As renewable energy generation increases, hours of oversupply regularly lead to negative electricity prices (where generators pay off-takers to consume power).



https://www.smard.de/home/marktdaten/

Electricity price forecasting (EPF)

Challenges in intra-day and day-ahead EPF include

- sudden, severe jump dynamics (spikes) in times when demand exceeds cheap generation, and high-cost peaking units (e.g., gas turbines) are triggered
- negative prices
- overlapping, complex seasonal patterns (intra-day, weekly, annual)
- structural breaks (regulatory interventions, fuel price shifts, etc.)
- dependence on exogenously uncertain features (weather forecasts)

Overviews of challenges, forecasting methods, and applications: Weron (2014); Nowotarski and Weron (2018)

Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.2.1 Economic forecasting

1.2.2 Political and social science

1.2.3 Sports analytics

1.2.4 Health and epidemiology

1.2.5 Energy forecasting

1.2.6 Weather and climate

1.2.7 Other environmental sciences applications

1.3 Forecasting competitions and prediction markets

Weather forecasts and their societal and economic importance

Weather forecasts have substantial economic and societal value:

- Public safety: early warnings to enable preparation for storms, floods, heatwaves and other hazardous events, saving lives and reducing damage.
- Economic decisions for weather-dependent industries such as agriculture, energy production and trading, transportation, construction, or insurance
- Operational planning to help allocate resources, schedule activities and manage weather-related risks
- Everyday life and personal decisions

Weather: Forecast data

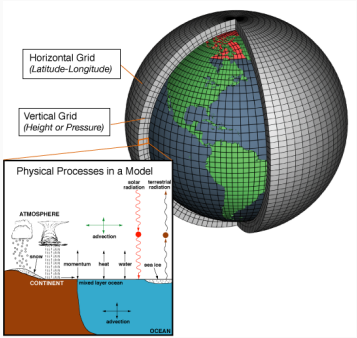


Figure: NOAA

Modern weather forecasts rely on physical models of the atmosphere.

Development of the state of the atmosphere on a 3D grid:

time t
initial conditions



physical laws
$\longrightarrow$
partial differential equations

time $t+1$
prediction



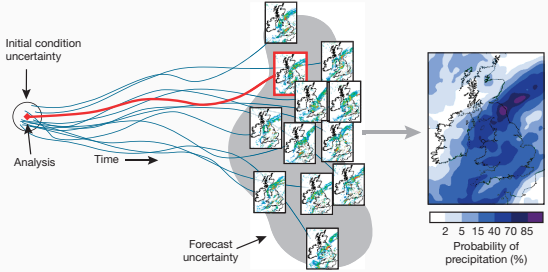
physical laws
$\longrightarrow$
partial differential equations

time $t+2$
prediction



Ensemble forecasts

Ensemble simulations allow for quantifying forecast uncertainties due to initial conditions, model errors, and non-linear error growth in a chaotic system.



Bauer et al. (2015)

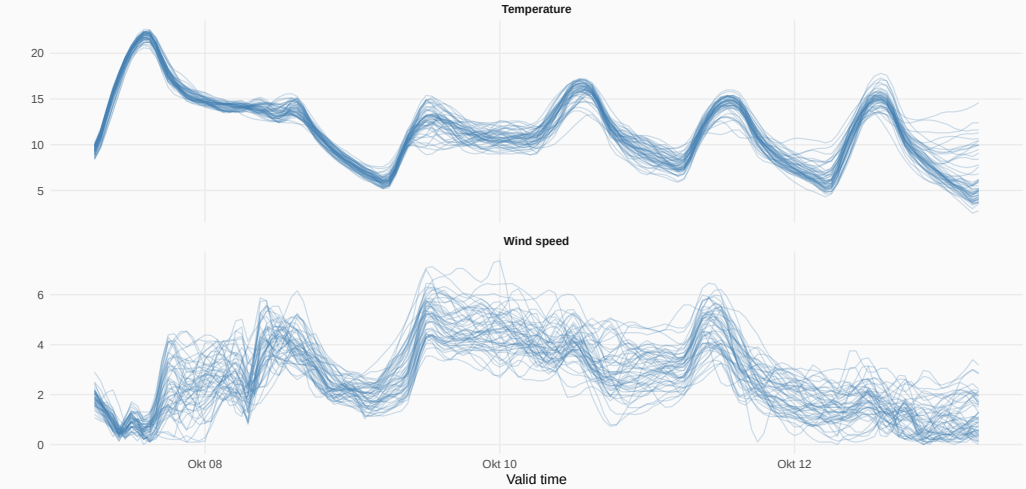


ECMWF

Overview, history, and further reading: Bauer et al. (2015)

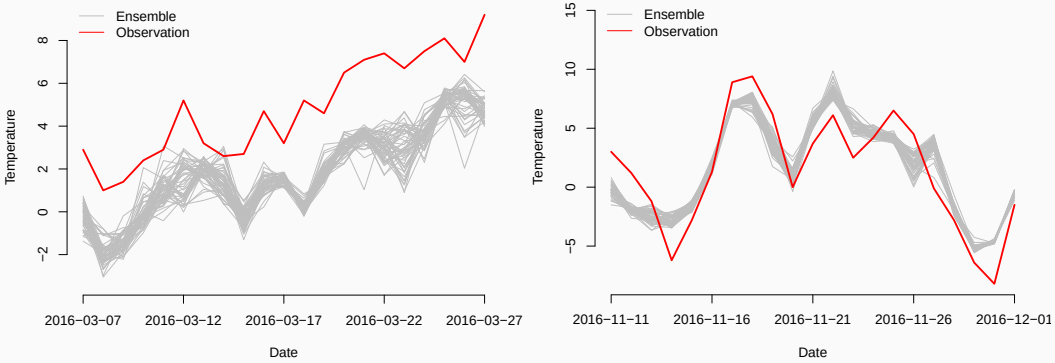
Ensemble forecasts

Exemplary ensemble forecasts for Frankfurt airport, initialized on October 7, 2026 at 06:00 (UTC)



Typical errors of ensemble weather forecasts

48-hour ahead ECMWF ensemble forecasts of 2-meter temperature at Frankfurt Airport, Germany



Post-processing ensemble weather forecasts

As ensemble forecasts are subject to biases and dispersion errors, some form of statistical post-processing is required.

Given ensemble forecasts of 2-meter temperature,

$$\boldsymbol{X}^{\text{t2m}} = (x_1, ..., x_M),$$

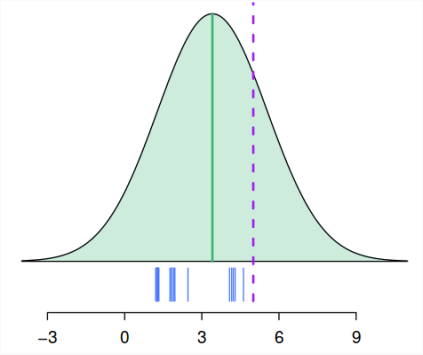
(parametric) post-processing approaches model the conditional distribution of the target variable y as a parametric distribution,

$$y \mid x_1, ..., x_M \sim F_\theta,$$

with the aim of correcting systematic errors by exploiting structure in past forecast-observation pairs.

Post-processing ensemble weather forecasts: EMOS

One popular approach is ensemble model output statistics (EMOS; Gneiting et al., 2005), where F_θ is a single parametric distribution. For example, for temperature forecasting the post-processed forecast takes the form of a Gaussian distribution.



$$y|\boldsymbol{X}^{\text{t2m}} \sim \mathcal{N}_{(\mu,\sigma)},$$

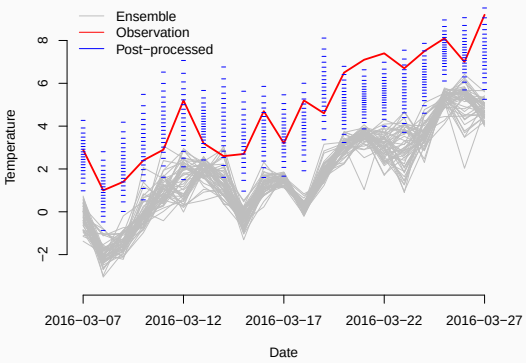
$$\mu = a + b \cdot \text{mean}(\boldsymbol{X}^{\text{t2m}})$$

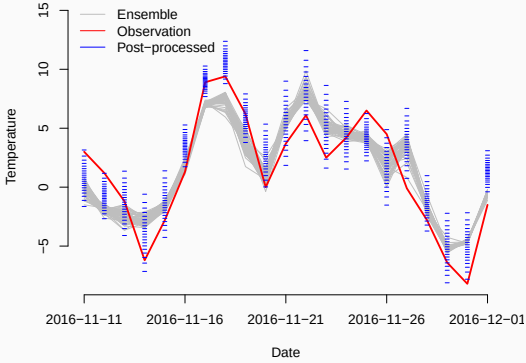
$$\sigma = c + d \cdot \text{sd}(\boldsymbol{X}^{\text{t2m}})$$

The EMOS coefficients a, b, c, d need to be estimated from training data.

Post-processing ensemble weather forecasts: EMOS

48-hour ahead ECMWF ensemble forecasts of 2-meter temperature at Frankfurt Airport, Germany





Post-processing ensemble weather forecasts: EMOS

If the ensemble forecasts $x_1, ..., x_M$ can not be considered exchangeable (e.g., if they are produced by different weather models), an alternative model formulation is

$$y \,|\, x_1, \ldots, x_M \sim \mathcal{N}(a_0 + a_1 x_1 + \cdots + a_M x_M,\ b_0 + b_1 s),$$

with bias parameters a_0, $a_1, \ldots, a_M$, and spread parameters b_0 and b_1, where s is the standard deviation of the ensemble values.

Post-processing ensemble weather forecasts: EMOS

EMOS specifications for continuous weather quantities:

Weather quantity	Range	Predictive distribution(s)
Temperature	$y \in \mathbb{R}$	Normal; skew-normal; mixtures of normals
Pressure	$y \in \mathbb{R}$	Normal
Precipitation amount	$y \in \mathbb{R}_+$	Left-censored GEV; censored, shifted gamma
	$y^{1/2} \in \mathbb{R}_+$	Left-truncated logistic
Wind components (u, v)	$(u, v) \in \mathbb{R}^2$	Bivariate normal
Wind speed	$y \in \mathbb{R}_+$	Left-truncated normal; left-truncated logistic; log-normal; GEV
Wind gusts	$y \in \mathbb{R}_+$	Left-truncated normal; left-truncated logistic; GEV

Post-processing ensemble weather forecasts: BMA

Bayesian model averaging (BMA; Raftery et al., 2005) employs a mixture distribution, where each component is a parametric density associated with an individual ensemble member.

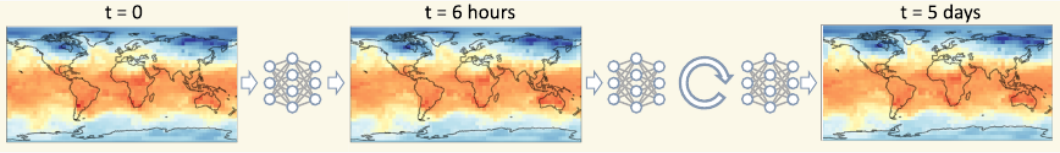
The mixture weight reflects the member's relative contributions to predictive skill over a training period.

For example, for temperature forecasting, the BMA mixture components can be chosen as Gaussian with a linearly bias-corrected mean, and so the BMA forecast density is

$$y \mid x_1, \ldots, x_M \sim \sum_{m=1}^{M} w_m \mathcal{N}(c_{0m} + c_{1m} x_m, \sigma^2),$$

with BMA weights, $w_1, \ldots, w_M$, that are nonnegative and sum to 1, bias parameters $c_{01}, \ldots, c_{0M}$ and $c_{11}, \ldots, c_{1M}$, and a common variance, σ^2.

AI-based data-driven weather prediction as a new paradigm



Rasp et al. (2024)

Potential benefits of AI weather models:

- better forecasts than physics-based weather models?
- potential of skillful forecasts at time scales or in locations where physical models have little or no skill
- substantially lower computational costs and faster generation of forecasts (once trained)

Do AI weather models produce better forecasts than physics-based models?

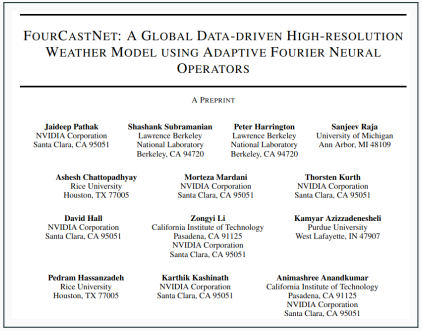
... probably (depending on the definition of *better*) ...

"History" of data-driven weather prediction

- early attempts (ca. 2018): proof of concept, but forecast skill is only comparable to simple toy models
- incremental progress, but forecast skill remains far behind that of operational physics-based models
- consensus (ca. 2021): challenging operational physics-based weather models seems highly unlikely for the foreseeable future

The AI revolution of weather prediction?

Unprecedented and surprising developments since February 2022:



FOURCASTNET: A GLOBAL DATA-DRIVEN HIGH-RESOLUTION WEATHER MODEL USING ADAPTIVE FOURIER NEURAL OPERATORS

A PREPRINT

Jaideep Pathak NVIDIA Corporation Santa Clara, CA 95051

Shashank Subramanian Lawrence Berkeley National Laboratory Berkeley, CA 94720

Peter Harrington Lawrence Berkeley National Laboratory Berkeley, CA 94720

Sanjeev Raja University of Michigan Ann Arbor, MI 48109

Ashesh Chattopadhyay Rice University Houston, TX 77005

Morteza Mardani NVIDIA Corporation Santa Clara, CA 95051

Thorsten Kurth NVIDIA Corporation Santa Clara, CA 95051

David Hall NVIDIA Corporation Santa Clara, CA 95051

Zongyi Li California Institute of Technology Pasadena, CA 91125 NVIDIA Corporation Santa Clara, CA 95051

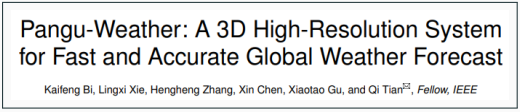
Kamyar Azizzadenesheli Purdue University West Lafayette, IN 47907

Pedram Hassanzadeh Rice University Houston, TX 77005

Karthik Kashinath NVIDIA Corporation Santa Clara, CA 95051

Animashree Anandkumar California Institute of Technology Pasadena, CA 91125 NVIDIA Corporation Santa Clara, CA 95051

Februar 2022: FourCastNet (Nvidia)



Pangu-Weather: A 3D High-Resolution System for Fast and Accurate Global Weather Forecast

Kaifeng Bi, Lingxi Xie, Hengheng Zhang, Xin Chen, Xiaotao Gu, and Qi Tian✉, *Fellow, IEEE*

November 2022: Pangu-Weather (Huawei)



GraphCast: Learning skillful medium-range global weather forecasting

Remi Lam[*,1], **Alvaro Sanchez-Gonzalez**[*,1], **Matthew Willson**[*,1], **Peter Wirnsberger**[*,1], **Meire Fortunato**[*,1], **Alexander Pritzel**[*,1], **Suman Ravuri**[1], **Timo Ewalds**[1], **Ferran Alet**[1], **Zach Eaton-Rosen**[1], **Weihua Hu**[1], **Alexander Merose**[2], **Stephan Hoyer**[2], **George Holland**[1], **Jacklynn Stott**[1], **Oriol Vinyals**[1], **Shakir Mohamed**[1] and **Peter Battaglia**[1]

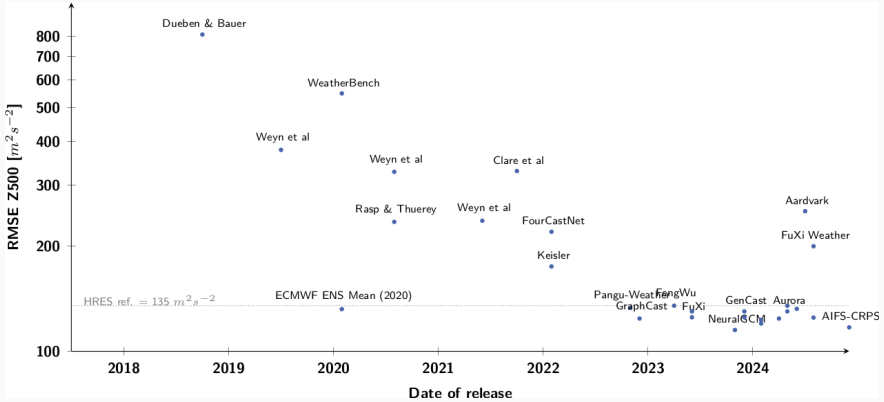
[*]equal contribution, [1]DeepMind, [2]Google

December 2022: GraphCast (Google/Deepmind)

... and more: ClimaX (Microsoft Research, January 2023), FengWu (Shanghai AI Laboratory, April 2023), FuXi (Fudan University, June 2023), ..., AIFS (ECMWF, June 2024), ...

Data-driven weather models: Successes

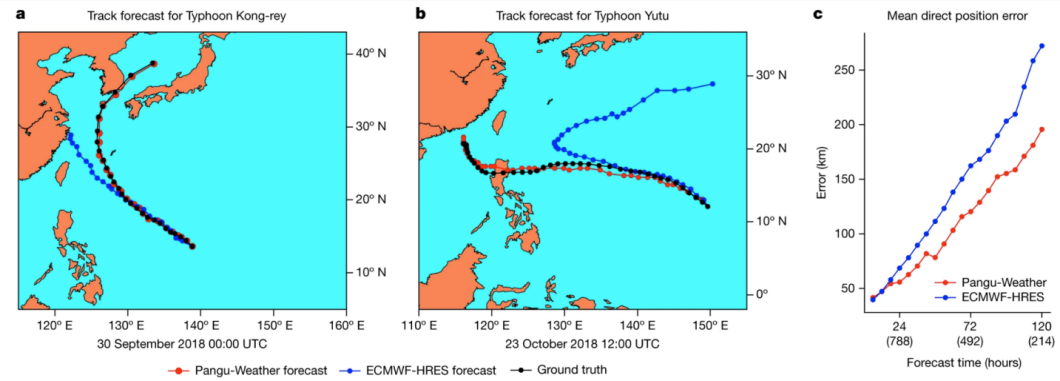
All these AIWP models reach or surpass the performance of state-of-the-art operational physical weather models ...



based on data collected by Stephan Rasp

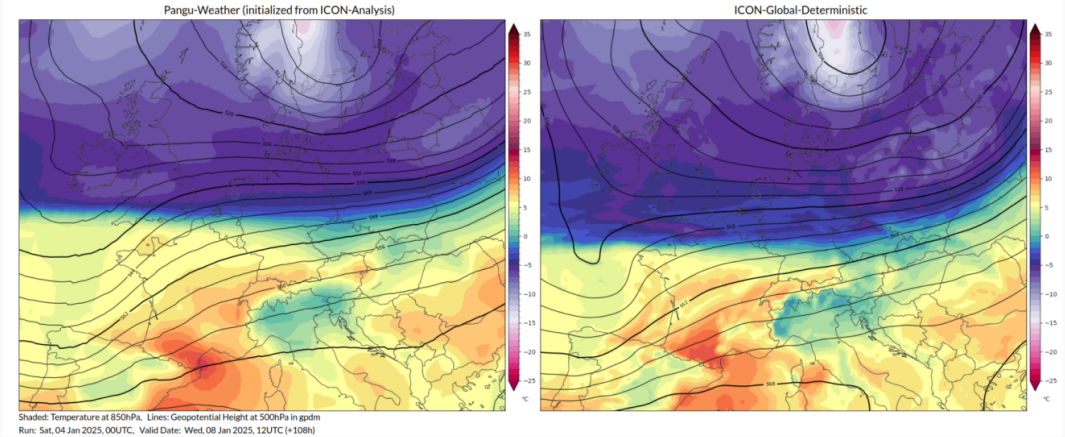
Data-driven weather models: Successes

... including predictions of complex phenomenons such as tropical cyclones



Bi et al. (2023), Nature

Exemplary Pangu-Weather forecasts



(Electricity) costs for one year of forecasts[1]

Physics-based NWP ensemble model



≈ 440 000 €

AI-based (Pangu-Weather) ensemble model



≈ 200 €

Training: ≈ 10 000 €

[1]Based on experiments on HoReKa for electricity costs of 50 Cent/kWh, 40 ensemble members, 2 models runs per day for 365 days. Images: generated by DALL-E 3 via Bing AI Copilot; lego.com

Climate forecasting

- Goal: Predict or project the future state and statistics of the Earth system (temperature, precipitation, extremes, etc.) over months to centuries:
  - Seasonal: months
  - Decadal: years–decades
  - Long-term: decades–centuries
- Approach:
  - Run a coupled Earth-system model (ESM) forward in time
  - Use ensembles to represent internal variability and model uncertainty
  - For long-term projections, specify future external forcings (e.g., greenhouse-gas and aerosol emissions)
- Outputs:
  - Probabilistic predictions of means, trends, and extremes
  - Climate indices (e.g., ENSO, NAO, PDO) and impact-relevant quantities

Weather vs. climate forecasting

- typical lead time: hours–weeks vs. months–centuries
- primary objective: predict the atmospheric state at a specific time vs. predict climate statistics and their evolution
- initialization: detailed atmospheric state vs. coupled Earth-system state; importance depends on lead time
- key physical processes: fast atmospheric dynamics vs. interactions between atmosphere, ocean, land, ice, and biosphere
- resolution: $\sim$1–10 km vs. $\sim$50–200 km
- typical users: aviation, emergency management, energy trading, ... vs. policy makers, impact modelers, long-term planners, ...

Challenges in climate forecasting

- Predictability: chaotic dynamics and intrinsic variability limit predictability, depending on the variable, spatial scale, and horizon
- Model uncertainty: unresolved physical processes (e.g., clouds, convection, ocean mixing) and imperfect parameterizations
- Initialization: incomplete knowledge of the current state of the coupled Earth system
- Forcing uncertainty: future greenhouse-gas emissions, aerosols, land use, and other external drivers are unknown
- Extremes: rare and potentially compound events are difficult to simulate and predict
- Evaluation: limited observations and few independent forecast cases, particularly for long lead times and extremes
- Uncertainty quantification: representing and communicating the different sources of uncertainty

Overview

1. Introduction, Examples and Applications

1.1 Probabilistic and point forecasts

1.2 Examples and applications

1.2.1 Economic forecasting

1.2.2 Political and social science

1.2.3 Sports analytics

1.2.4 Health and epidemiology

1.2.5 Energy forecasting

1.2.6 Weather and climate

1.2.7 Other environmental sciences applications

1.3 Forecasting competitions and prediction markets

Examples of environmental sciences applications and prediction targets

- Hydrology and water resources: river discharge, water levels, soil moisture, ...
- Air quality and atmospheric composition: pollutants such as PM2.5, ozone etc.
- Wildfires: ignition probability, fire occurrence, burned area, fire spread, smoke exposure, ...
- Other environmental hazards: earth quakes, landslides, avalanches, ...
- Oceanography and marine ecosystems: ocean currents, sea level, waves, sea ice, harmful algal blooms, ...
- Agriculture: crop yield, soil moisture, plant diseases, pest outbreaks
- Biodiversity and ecological dynamics: species abundance, occupancy, migration, population dynamics, ...

Examples of environmental sciences applications and prediction targets

These fields strongly differ in terms of the quantities of interest, forecast horizons, and the degree of maturity.

Forecasting methods and approaches often include combinations of

- process-based models (e.g., physics-based hydrological models or biological models of plants)
- data-driven approaches relying on observation data (e.g., measurements, satellite imagery, weather forecasts, climate model projections, ...)
- statistical and machine learning methods (time series models, Bayesian statistics, post-processing, spatio-temporal statistics, ...)
- hybrid approaches combining process-based and data-driven modeling

Hydrological forecasting

Floods are among the most dangerous natural hazards in Europe.

For example, the flooding in the Ahr valley and at the Erft in 2021 caused at least 188 deaths and economic losses of more than €40 billion.



https://www.bundeswirtschaftsministerium.de/Redaktion/DE/Downloads/M-O/Merkblaetter/merkblatt-klimawandelfolgen-in-deutschland-05.pdf

Hydrological forecasting

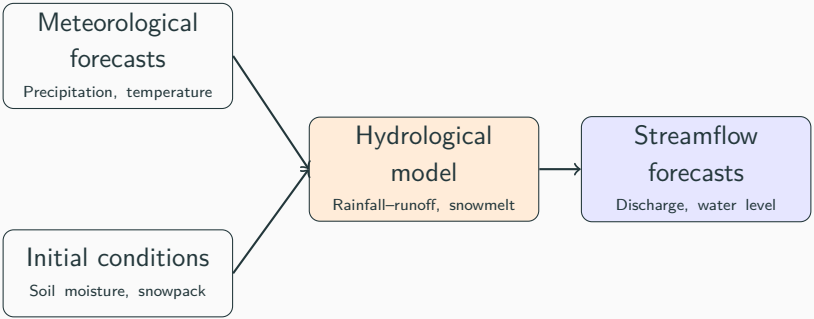
The absence of water can also cause severe economic problems.

In 2026, rivers in Germany experienced extremely low water levels, with economic damages estimated to be up to €2 billion, and decreasing GPD growth by up to 0.5 percentage points.



https://www.bafg.de/DE/5_Informiert/2_Publikationen/Niedrigwasserbericht/niedrigwasserbericht_node.html

Hydrological forecasting



Hydrological models

Conceptual water balance models (e.g., LARSIM) are often used by the flood forecasting centres. They divide the river basins into small computational units and simulate the main processes of the terrestrial water cycle, and the routing of water through the river network.

Observed meteorological data (station measurements and radar-based precipitation) drive the model up to the present. From there, numerical weather predictions, drive the model into the future and yield forecasts of discharge and water level.

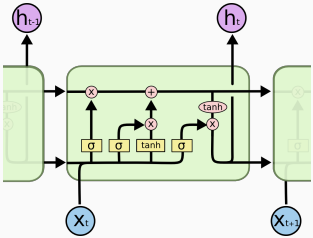
To account for uncertainties, these models are often run as ensembles, e.g., by utilizing ensemble weather forecasts as inputs.

Details and overviews: Cloke and Pappenberger (2009) and Troin et al. (2021).

Hydrological models

Over the past years, Long Short-Term Memory (LSTM) networks turned out to be exceptionally well suited for rainfall-runoff modeling.

Trained jointly on hundreds of catchments, LSTMs learn hydrological behavior that generalizes across basins and outperform established conceptual and process-based models.

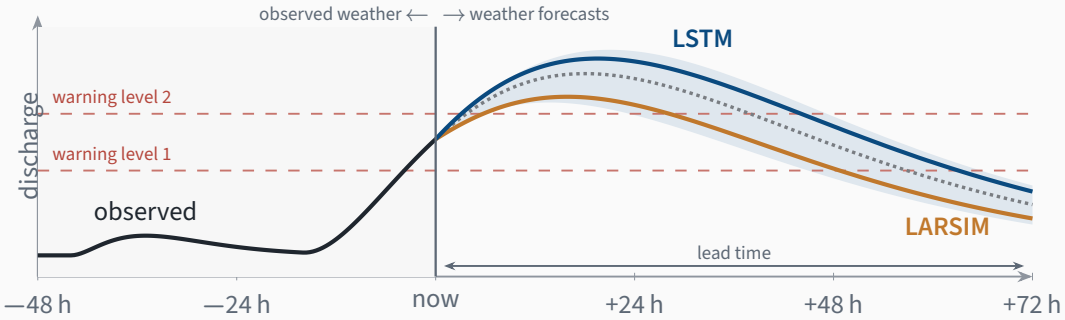


https://colah.github.io/posts/2015-08-Understanding-LSTMs

Google's Flood Hub issues operational LSTM-based flood forecasts in more than 80 countries.

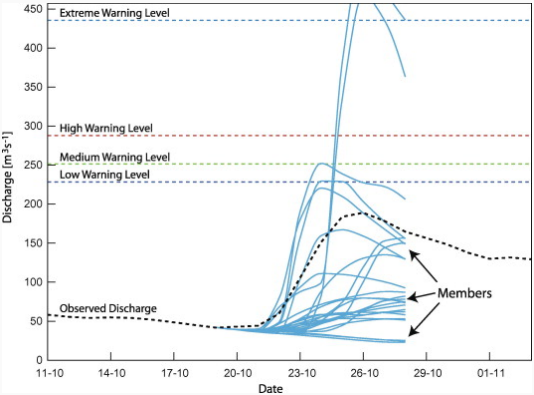
Details: Kratzert et al. (2018) and Nearing et al. (2024).

Hydrological forecasting



Hydrological ensemble models

Ensemble forecasts can be generated by using different hydrological models or ensemble weather forecasts



Cloke and Pappenberger (2009)

Post-processing can be applied to the ensemble weather forecasts used as inputs, but also to the hydrological ensemble forecasts obtained as output of the hydrological model.

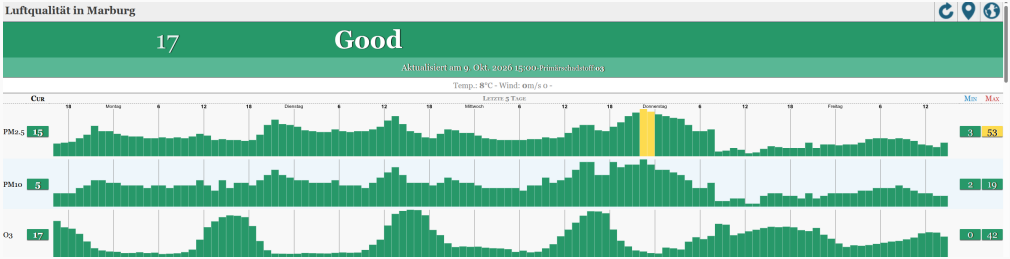
Advertisement

Advanced software lab / project work in the current winter semester: Development of an LSTM model for flood prediction in Hesse, together with HLNUG.

Application is possible until October 25, 2026.



Air quality forecasting



https://aqicn.org/city/germany/hesse/marburg/de/m/

Central question: How likely is the concentration of a pollutant to exceed a health-relevant threshold, at a particular location and time? Target variables:

- particulate matter (PM10, PM2.5) from traffic, industry, domestic heating, wildfires
- gaseous pollutants (O_3, NO_2, SO_2, CO) from atmospheric chemical reactions (ozone), traffic, combustion, and industry (others)
- smoke and dust from wildfires or Sahara dust episodes

Air quality forecasting

Emissions and observations

Traffic, industry, fires; monitoring stations and satellites

↓

Meteorological forecasts

Wind, temperature, boundary-layer mixing, precipitation

↓

Chemical transport model

Transport, dispersion, chemical reactions, deposition

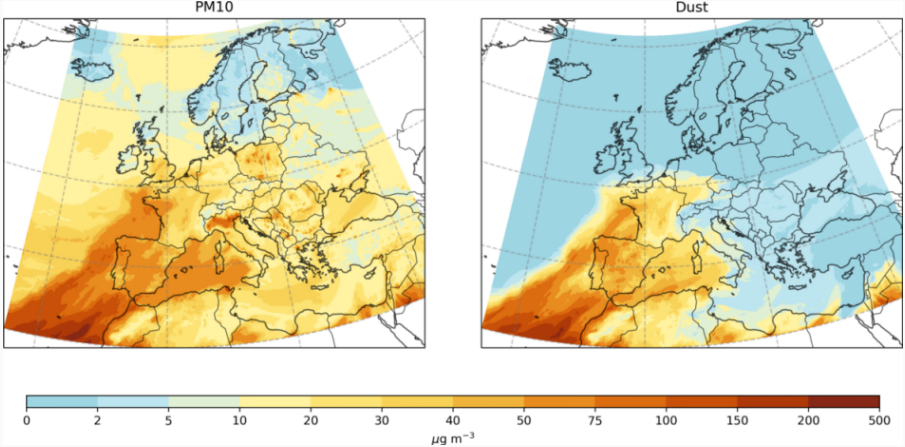
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Forecast and uncertainty estimation

Multi-model ensemble, statistical post-processing, predictive distributions

Air quality forecasting

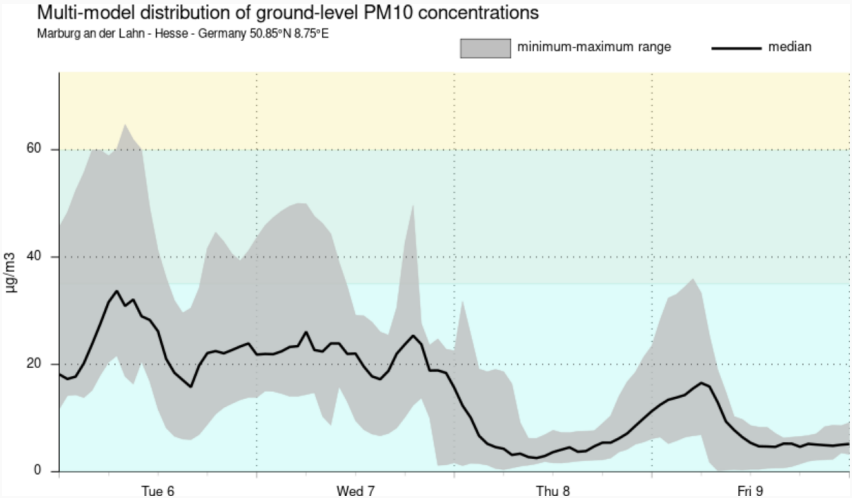
The Copernicus Atmosphere Monitoring Service (CAMS) provides regional air-quality ensemble forecasts from 11 chemical transport models



https://atmosphere.copernicus.eu/cams-air-boreal-winter-2324

Air quality forecasting

Real-time forecasts: https://charts.ecmwf.int/catalogue/packages/cams_air_quality_eea/



Overview

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1.3 Forecasting competitions and prediction markets

Forecasting competitions

Structured, open experiments in which participants submit predictions for a common set of future outcomes, and performance is evaluated using a predefined metric.

Different types of competitions:

- domain-specific, statistical and machine-learning forecasting competitions
- general purpose predictive modelling competitions (e.g., on Kaggle), often focusing on classification or regression rather than forecasting future events
- human geopolitical and current-events forecasting tournaments
  - examples: Metaculus, Good Judgement Project[2]

[2] see also Tetlock, P.E. (2005) *Expert Political Judgment: How Good Is It? How Can We Know?*

Forecasting competitions: Examples

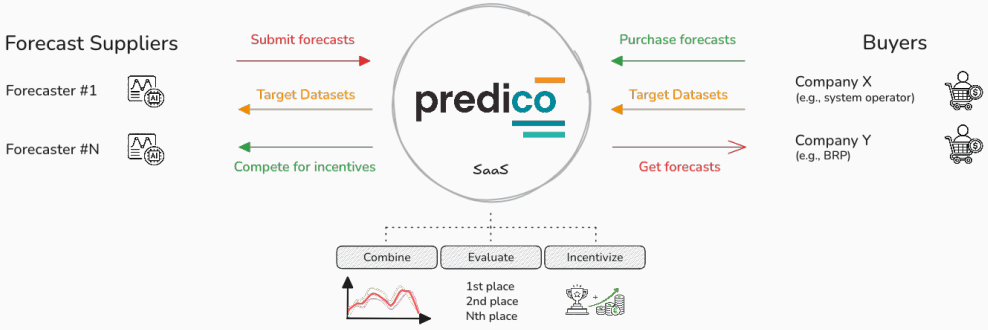
- Global Energy Forecasting Competition (GEFCOM) 2012, 2014, 2017 with a focus on load, electricity price, wind power and solar power forecasting
- Makridakis (or M-)Competitions since 1982 (latest edition M6 Competition 2022) for time series forecasting
- WMO Sub-seasonal to Seasonal (S2S) AI Challenge on sub-seasonal weather prediction

Real-time forecasting competitions:

- Survey of Professional Forecasters, macroeconomic forecasting
- COVID-19 Forecasting Hub, epidemiology, `https://covid19forecasthub.org/`
- ECMWF AI Weather Quest, subseasonal weather prediction, `https://aiweatherquest.ecmwf.int/`

Collaborative forecasting platforms with market structures

Platforms such as Predico[3] are multilateral, collaborative markets for forecasts, connecting entities with common interests to procure and sell forecasts.

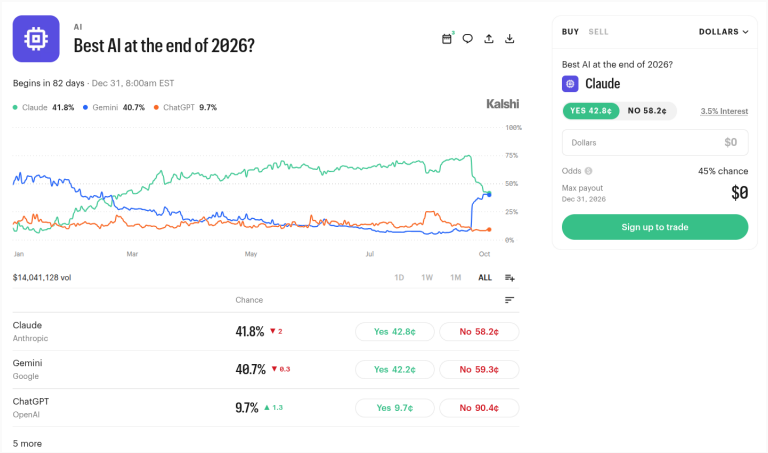


https://predico.inesctec.pt/docs/

[3] https://innovation.eliagroup.eu/en/projects/predico-collaborative-forecasting-platform

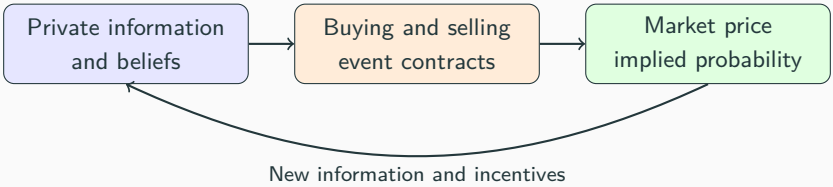
Prediction markets

Prediction markets allow participants to buy and sell contracts whose payoffs depend on the outcome of a future event. The market price aggregates information through participants' willingness to trade.



https://kalshi.com/markets/kxllm1/yearend-top-llm/kxllm1-26dec31

Prediction markets



- A binary contract pays 1 if an event occurs and 0 otherwise.
- Its price can be interpreted as an implied probability.
- Trading aggregates information, but prices need not be calibrated.
- Examples: Polymarket, Kalshi, Manifold Markets.

Prediction markets

While proponents argue that prediction markets aggregate information efficiently to generate accurate real-time forecasts, these markets raise several major ethical concerns:

- insider trading and manipulation without a regulatory framework
- incentives to cause predicted events
- gambling under the guise of finance
- gamification and aggressive social media marketing of prediction platforms disproportionately targets younger traders
- profiting from misfortune and human suffering by betting on negative events
- predictions might change outcomes in reality, e.g., panic selling in stock markets or alterations in political strategy

Further reading: `https://cer.econ.columbia.edu/news/use-caution-hidden-risks-prediction-markets`

Insider trading on prediction markets

- Between 30 December 2025 and 2 January 2026, a US special forces soldier places a total of $32,500 on bets on Venezuela's President Nicolás Maduro being out of office by the end of January 2026.

  When Maduro is seized by US special forces on 3 January 2026, the soldier wins $436,000.

- Surrounding the US strikes on Iran in February 2026, six accounts created on Polymarket in February win $1.2 million in wagers on a US strike on Iran happening by 28 February.

  One of those accounts later made $163,000 by correctly betting on a US-Iran ceasefire by 7 April.


BBC

Home News Sport Business Technology Health Culture Arts Travel Earth Audio Video Live

The insider trading suspicions looming over Trump's presidency

20 April 2026

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https://www.bbc.com/news/articles/cge0grppe3po